PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Spin within systematic review abstracts on antiplatelet therapies after acute coronary syndrome: A cross-sectional study
AUTHORS	Wise, Audrey; Mannem, Deepika; Arthur, Wade; Ottwell, Ryan; Greiner, Benjamin; Srouji, Derek; Wildes, Daniel; Hartwell, Micah; Wright, Drew; Khojasteh, Jam; Vassar, Matthew

VERSION 1 – REVIEW

REVIEWER	Naichuan Su Academic Centre for Dentistry Amsterdam
REVIEW RETURNED	14-Jun-2021

GENERAL COMMENTS	The study assessed the spin In the systematic review abstracts on antiplatelet therapies after acute coronary syndrome. The topic is very interesting and popular recently in research and the authors did nice work on it. However, I have some comments as below: 1. In the study, the authors only included the 9 types of spin which are most severe based on Yavchitz et al. The authors are advised to explain in the Methods section how the “severe” spin types were defined and selected. 2. Based on Yavchitz et al, only the first 8 spin types had a mean rating of severity of >0 and the 9th spin type had the mean rating of severity of <0. Why the authors also included the 9th spin type when it was reported to be less severe? 3. Page 12 (line 26-31), the authors used binary logistic regression analysis to determine the association between different factors and occurrence of spin. This means that the spin of the study, which is the outcome variable in the regression analysis, is dichotomized. The authors are advised to clarify how the spin in the studies was dichotomized in the methods section. 4. In the conclusion section (both in abstract and the main text), the authors are advised to also mention there is no significant association identified between the occurrence of spin and the factors.

REVIEWER	Rohan Magoon Dr Ram Manohar Lohia Hospital PGIMER, Cardiac Anaesthesia
REVIEW RETURNED	16-Sep-2021

GENERAL COMMENTS	Many congratulations on the index research endeavor as elucidating potential "spin" in literature is the need of hour. At the same time, mitigation of the same is equally important, be it in the present context of antiplatelet therapy in ACS or in general, as the consequences of overlooking it can be long-lasting.

	While the idea and presentation style is impressive, I would suggest the authors to discuss a few more important aspects about common "spin-motivators" in research. Suggestions to deescalate spin need to be elaborated in the light of recent relevant literature, as evidenced in: Avidan MS, Ioannidis JPA, Mashour GA. Independent discussion sections for improving inferential reproducibility in published research. Br J Anaesth. 2019 Apr;122(4):413-420. doi: 10.1016/j.bja.2018.12.010. Epub 2019 Jan 22. PMID: 30857597; PMCID: PMC6435840. This becomes important as "inferential-reproducibility" is also an important consideration. Reading the article: Safeguarding anaesthesia research from spin. Br J Anaesth. 2020 Dec;125(6):e460-e462. doi: 10.1016/j.bja.2020.08.042. can assist the authors in revising their discussion. Kindly appropriately cite the above mentioned as and when referred to. Best wishes,,,

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Naichuan Su, Academic Centre for Dentistry Amsterdam

Comments to the Author:
The study assessed the spin In the systematic review abstracts on antiplatelet therapies after acute coronary syndrome. The topic is very interesting and popular recently in research and the authors did nice work on it. However, I have some comments as below:

1. In the study, the authors only included the 9 types of spin which are most severe based on Yavchitz et al. The authors are advised to explain in the Methods section how the “severe” spin types were defined and selected.

Authors' reply: Thank you for bringing this topic to our attention. We have changed the text in our manuscript to address this comment. The following was added to the manuscript:

“These forms of spin as defined in Yavchitz et al.(Yavchitz et al. 2016) studies were chosen and settled on following a Q-sort survey with members of the Cochrane Collaboration. This consisted of method review and bias method group members from over 400 invited participants. Members ranked examples on the Q-sort survey for more severe and prevalent forms of bias with the top 9 being evaluated in our study. “

2. Based on Yavchitz et al, only the first 8 spin types had a mean rating of severity of >0 and the 9th spin type had the mean rating of severity of <0. Why the authors also included the 9th spin type when it was reported to be less severe?

Authors' reply: Although not as severe, we chose the top 9 for completeness given our study sought to apply the same research methodology and design as the Yavchitz et al. study . We have added the following sentence in the text to address this comment:

The selection of the 9 types of spin in our study is derived from the most severe types of spin Yavchitz el al(Yavchitz et al. 2016) identified as well as from recent published literature(Ottwell et al. 2020) on spin for completeness.

3. Page 12 (line 26-31), the authors used binary logistic regression analysis to determine the association between different factors and occurrence of spin. This means that the spin of the study, which is the outcome variable in the regression analysis, is dichotomized. The authors are advised to clarify how the spin in the studies was dichotomized in the methods section.

Authors' reply: Thank you for requesting clarification within our methodology as we always strive for increased reproducibility and transparency. The following has been added to the manuscript within the methods section:

Researchers coded each of the 9 spin types as being either present or absent from the systematic review, except for type 4 spin if safety was not addressed in the manuscript. If a systematic review did not address safety, it was not included in the analysis for type 4 spin.

4. In the conclusion section (both in abstract and the main text), the authors are advised to also mention there is no significant association identified between the occurrence of spin and the factors.

Authors' reply: Thank you for your comment to insert the lack of significant associations in our study. The following is found in the abstract and conclusion:

abstract:
There were no statistically significant associations between spin and study characteristics or AMSTAR-2 ratings; however, implementing changes will ensure that spin is reduced in the field of cardiology as well as other fields of medicine.

conclusion:
Although there were no statistically significant relationships between spin and the evaluated study characteristics or AMSTAR-2 appraisals, this misrepresentation of research data poses a threat to optimal evidence-based clinical use of antiplatelet therapy following ACS.

Reviewer: 2
Dr. Rohan Magoon, Dr Ram Manohar Lohia Hospital PGIMER

Comments to the Author:
Many congratulations on the index research endeavor as elucidating potential "spin" in literature is the need of hour.

At the same time, mitigation of the same is equally important, be it in the present context of antiplatelet therapy in ACS or in general, as the consequences of overlooking it can be long-lasting.

While the idea and presentation style is impressive,
I would suggest the authors to discuss a few more important aspects about common "spin-motivators" in research.

Authors' reply: Thank you for this comment. It is important to express why a motivation for spin in scientific research exists. We have changed the following in our manuscript:

Whether intentional or not, motivating factors for the use of spin within manuscripts include confirmation bias, allegiance bias, and the misrepresentation of results in hopes for publication. Ioannidis et al suggest that the motives include a belief that positive results are associated with an increased likelihood of publication and greater clinical significance (Independent discussion sections for i...).

Suggestions to deescalate spin need to be elaborated in the light of recent relevant literature, as evidenced in: Avidan MS, Ioannidis JPA, Mashour GA. Independent discussion sections for improving inferential reproducibility in published research. Br J Anaesth. 2019 Apr;122(4):413-420. doi: 10.1016/j.bja.2018.12.010. Epub 2019 Jan 22. PMID: 30857597; PMCID: PMC6435840.

Authors' reply: Thank you for the recommended citation to strengthen our discussion. We agree the deescalation of spin is vital and we have included the following in our manuscript:

Furthermore, in an effort to deescalate spin, we urge journal editors to strive for inferential reproducibility by using a second discussant to verify study findings prior to publication(Independent discussion sections for i...). Additionally, we recommend journal editors emphasize the publication of studies with neutral results given that researchers believe positive results are more likely to be published (Magoon and Jose 2020).

Given that Bero et al identified that spin is less prevalent in publications evaluating spin(Bero et al. 2019), we believe an awareness of spin is vital to decreasing spin in publications. Thus, by creating an online resource dedicated to teaching spin, as well as presenting on spin at medical conferences, we believe knowledge of spin would be increased among medical students, residents, and physicians and its prevalence in publications may decrease.

This becomes important as "inferential-reproducibility" is also an important consideration. Reading the article: Safeguarding anaesthesia research from spin. Br J Anaesth. 2020 Dec;125(6):e460-e462. doi: 10.1016/j.bja.2020.08.042. can assist the authors in revising their discussion.

Authors' reply: Thank you for the recommendation to include inferential reproducibility in our discussion. The following statement, alongside the previous citation, has been added to our discussion. Below is our statement defining inferential reproducibility:

Inferential reproducibility is attained for a publication by an independent reviewer re-analyzing the study to see if the same results and conclusions are reached (Magoon and Jose 2020).